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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	GEEK FAM
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4 & 8
Project / Solution Name	AI Career Navigator
Name (Leader)	Muhammad Ammar Hariz Bin Yazid Hamdi
Email & Phone (Leader)	ammarhariz140807@gmail.com / 01139060828
University Affiliation(s)	Universiti Teknikal Malaysia Melaka (UTeM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Many university students and fresh graduates in Malaysia struggle to choose a suitable career path because they lack clear, personalized guidance on their interests, skills and the jobs available. As a result, many graduate unsure which skills employers expect and how to prepare for them. According to the Department of Statistics Malaysia (DOSM), skill-related underemployment among employed persons with tertiary education stood at 35.2% in the second quarter of 2026, meaning more than one in three are working in jobs below their qualifications.

The problem is growing as artificial intelligence (AI) and automation change the skills required across industries. Without proper guidance, students may spend time and money learning skills that do not match their career goals or miss opportunities in emerging fields. This leads to skills mismatch, underemployment, lower confidence and a slower transition from education to work. It also affects employers who cannot find suitably skilled graduates, and the wider economy.

The people most affected are university students, fresh graduates and job seekers with limited access to personalized career counselling. The challenge connects directly to SDG 4: Quality Education, through the learning needs students must identify, and SDG 8: Decent Work and Economic Growth, through employability and job readiness.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AI Career Navigator is a web-based application designed to help students and job seekers make more informed career decisions through personalized AI-powered guidance. Users can provide information about their interests, existing skills, educational background, and career goals to receive recommendations that match their profiles.

The application uses a Generative AI model to interpret users' information, identify potential skill gaps, suggest suitable career paths, and generate personalized learning recommendations. It can also provide practical next steps, such as skills to prioritize, learning topics to explore, and ways to prepare for future employment. The aim is to turn general career advice into a more structured and actionable plan.

The proposed solution combines a user-friendly web interface, an AI service, and a database to process user inputs and manage relevant information. Cloud hosting will allow users to access the application online without installing additional software.

During the SDC Phase 1 workshops, our team intends to apply the skills gained in Generative AI API integration, database architecture, and cloud deployment to develop a functional prototype. We will focus on making the recommendations understandable, relevant, and practical for students and job seekers.

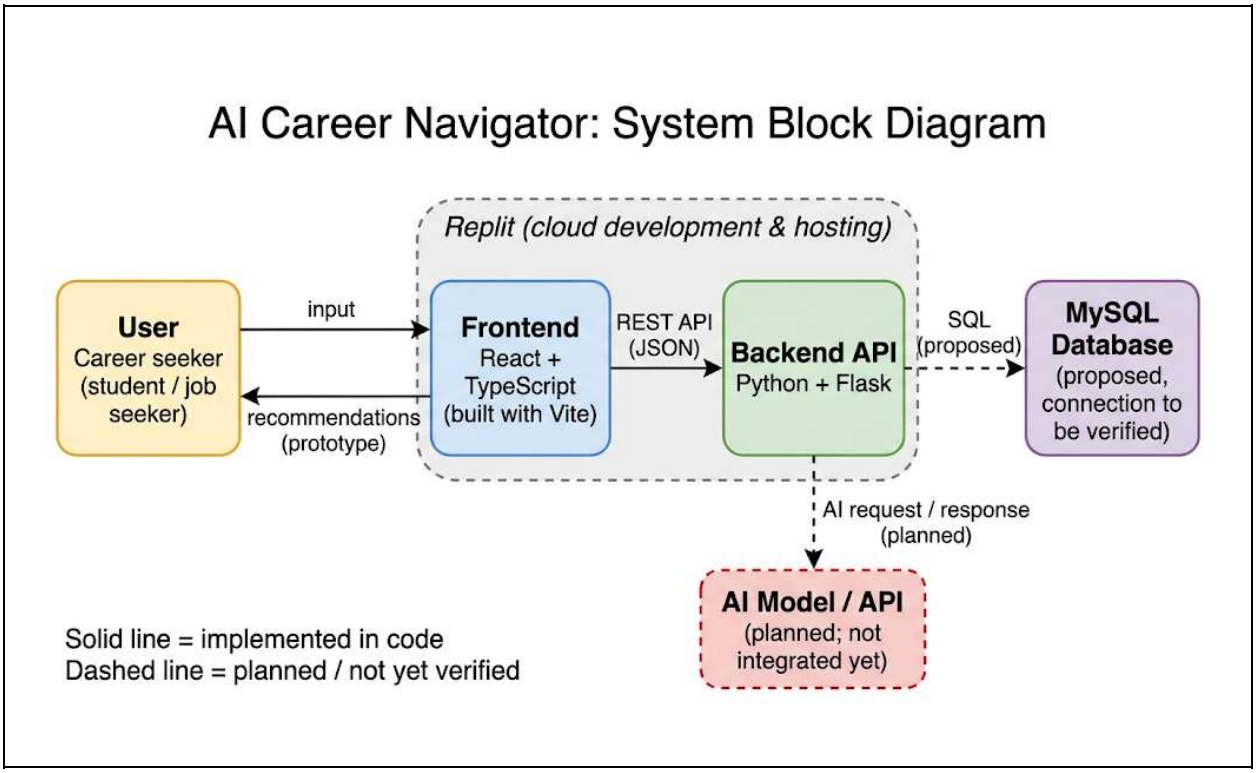
By improving access to personalized career guidance, AI Career Navigator aims to support informed career planning, encourage continuous learning, and help users prepare for changing workforce demands.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We followed the five stages of design thinking, adapted for a student-built prototype.

Empathize: As university students, we see that many students feel unsure about which career path to choose.

Define: We defined the problem as the lack of simple, personalised career guidance for students and job seekers.

Ideate: We decided on a web application where users enter their profile and receive career suggestions.

Prototype: We built a working web prototype using React, TypeScript and Flask on Replit, and deployed it online at https://ai-career-navigator-ayiqq.replit.app/.

Test (planned): In Phase 2, we will test the app with real students, collect feedback on ease of use, and improve the interface.

Deployment plan: Before Phase 2, we will verify the MySQL database connection, integrate a real AI model through the backend, and test the system for reliability. Since the app is already hosted on Replit, updates can be redeployed quickly.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s)	**SDG Target 8.6: "substantially reduce the proportion of youth not in employment, education or training" (Indicator 8.6.1, youth aged 15–24). AI Career Navigator supports this by giving young people easier career guidance, so they can move towards jobs or training.**
Current Problem Baseline	**In August 2026, Malaysia's youth unemployment rate (aged 15–24) was 10.1%, about 291,300 people, compared with 2.9% for the whole country (DOSM, Labour Force Statistics August 2026).**
Core Measurable KPI	Percentage of pilot users who report clearer career direction after using the app, measured with a short feedback survey (rating 4 or 5 out of 5).
Phase 2 Impact Target	By the Phase 2 presentation, at least 70% of pilot testers (target group: at least 30 UTeM students) report clearer career direction.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code (frontend library)	MIT
TypeScript	Code (programming language/compiler)	Apache-2.0
Vite	Code (build tool)	MIT
Flask	Code (backend framework)	BSD-3-Clause
MySQL Community Edition	Software (database)	GPL-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

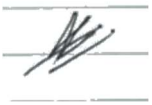
☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

None

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Muhammad Ammar Hariz Bin Yazid Hamdi]

[24/10/2026]